

# HDOC Day-27 - Complete Solution

**Topic:** Menu-driven C program using switch statement, nested loops, arrays, and separate parameterless functions.

**Note:** Each pattern function takes the number of rows from the user internally, so the functions are parameterless and return no value.

## General Algorithm

| Step | Algorithm                                                                                 |
|------|-------------------------------------------------------------------------------------------|
| 1    | Start the program.                                                                        |
| 2    | Display a menu containing the five pattern choices and Exit.                              |
| 3    | Read the user's choice.                                                                   |
| 4    | Use a switch statement to call the corresponding parameterless function.                  |
| 5    | Inside the selected function, input n and use nested loops to print the required pattern. |
| 6    | Return to the menu until the user selects Exit.                                           |
| 7    | Stop.                                                                                     |

## Test-Case Table

| Test | Input         | Expected result              |
|------|---------------|------------------------------|
| 1    | Choice=1, n=4 | Full pyramid with 4 rows     |
| 2    | Choice=2, n=4 | Inverted pyramid with 4 rows |
| 3    | Choice=3, n=4 | Diamond with 7 rows (2n-1)   |
| 4    | Choice=4, n=4 | Palindromic number pyramid   |
| 5    | Choice=5, n=4 | Repeated-number pyramid      |
| 6    | Choice=6      | Program exits                |
| 7    | Choice=9      | Displays Invalid choice      |

## C Program

```
#include <stdio.h>

void fullPyramid(void);
void invertedPyramid(void);
void diamondPattern(void);
void palindromicPyramid(void);
void repeatedNumberPyramid(void);

int main(void)
{
    int choice;

    do
    {
        printf("\n--- PYRAMID / DIAMOND MENU ---\n");
        printf("1. Full pyramid\n");
        printf("2. Inverted pyramid\n");
        printf("3. Diamond pattern\n");
        printf("4. Palindromic number pyramid\n");
        printf("5. Repeated-number pyramid\n");
        printf("6. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1: fullPyramid(); break;
            case 2: invertedPyramid(); break;
            case 3: diamondPattern(); break;
            case 4: palindromicPyramid(); break;
            case 5: repeatedNumberPyramid(); break;
            case 6: printf("Program ended.\n"); break;
            default: printf("Invalid choice! Try again.\n");
        }
    } while (choice != 6);

    return 0;
}

void fullPyramid(void)
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n - i; j++)
            printf(" ");
        for (j = 1; j <= 2 * i - 1; j++)
            printf("* ");
        printf("\n");
    }
}

void invertedPyramid(void)
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = n; i >= 1; i--)
    {
        for (j = 1; j <= n - i; j++)
            printf(" ");
        for (j = 1; j <= 2 * i - 1; j++)
            printf("* ");
        printf("\n");
    }
}

void diamondPattern(void)
{
    int n, i, j;
    printf("Enter number of rows for upper half: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
```

```

        for (j = 1; j <= n - i; j++)
            printf(" ");
        for (j = 1; j <= 2 * i - 1; j++)
            printf("* ");
        printf("\n");
    }

    for (i = n - 1; i >= 1; i--)
    {
        for (j = 1; j <= n - i; j++)
            printf(" ");
        for (j = 1; j <= 2 * i - 1; j++)
            printf("* ");
        printf("\n");
    }
}

```

```

void palindromicPyramid(void)
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n - i; j++)
            printf(" ");

        for (j = 1; j <= i; j++)
            printf("%d ", j);

        for (j = i - 1; j >= 1; j--)
            printf("%d ", j);

        printf("\n");
    }
}

```

```

void repeatedNumberPyramid(void)
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n - i; j++)
            printf(" ");

        for (j = 1; j <= 2 * i - 1; j++)
            printf("%d ", i);

        printf("\n");
    }
}

```

## Expected Patterns

Sample outputs for n = 4

1) Full pyramid

```
  *
 * * *
* * * * *
* * * * * * *
```

2) Inverted pyramid

```
* * * * *
 * * * *
  * * *
   *
    *
```

3) Diamond pattern

```
  *
 * * *
* * * * *
* * * * * * *
 * * * * *
  * * *
   *
    *
```

4) Palindromic number pyramid

```
  1
 1 2 1
1 2 3 2 1
1 2 3 4 3 2 1
```

5) Repeated-number pyramid

```
  1
 2 2 2
 3 3 3 3
 4 4 4 4 4 4
```

Concepts used: functions, switch-case, do-while loop, nested for loops, input/output using scanf and printf.